

5 **Draft Standard for**
6 **Local and metropolitan area networks—**

7 **Link Aggregation**

8 **Amendment 1:**
9 **YANG for Link Aggregation**

10 Prepared by the

11 **Time-Sensitive Networking (TSN) Task Group of IEEE 802.1**

12 Sponsor

13 **LAN/MAN Standards Committee**

14 **of the**

15 **IEEE Computer Society**

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17 for IEEE 802.1 Working Group members and will be updated as convenient. **New participants: Please read**
18 **these cover pages**, they contain information that should help you contribute effectively to this standards
19 development project. The [Introduction to the current draft](#) should be useful to all readers.

20 The text proper of this draft begins with the [Title page](#).

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28 Working Group and Time-Sensitive Networking (TSN) Task Group.

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1 **PAR (Project Authorization Request) and CSD**

2 This page is a draft, based on the proposed PAR and CSD as of the close of the May 2023 802.1 Interim
3 Meeting.

4 Extracts from the PAR, as approved by IEEE NesCom February 15, 2024:

5 <https://development.standards.ieee.org/myproject-web/public/view.html#pardetail/10885>

6 and the CSD (Criteria for Standards Development):

7 <https://mentor.ieee.org/802-ec/dcn/23/ec-23-0238-00-ACSD-p802-1axdz.pdf>

8 follow.

9 **Scope of the project:**

10 This amendment specifies YANG modules that allows configuration and status reporting for systems
11 implementing Link Aggregation, and optionally Distributed Resilient Network Interconnect, based on the
12 capabilities currently specified in clause 7 (management) and Annex D (Management Information Base
13 definitions). This amendment also includes technical and editorial corrections in the description of existing
14 IEEE Std 802.1AX functionality.

15 **PAR Need for the Project:**

16 YANG (IETF RFC 7950) is a formalized data modeling language that is widely accepted and can be used to
17 simplify network configuration. The ability to manage Link Aggregation via YANG modules is needed for
18 compatibility with modern network management systems.

19 **CSD broad market potential [extract]:**

20 The proposed amendment will support the use of YANG, which has broad industry support in networks that
21 use IEEE Std 802.1AX. Both IEEE Std 802.1AX and YANG are already supported and used by multiple
22 vendors, network providers, and network users. There is a wide interest in the industry to manage Link
23 Aggregation via YANG.

24 **Economic feasibility [extract]:**

- 25 a) Management using YANG utilizes a balance between end station and infrastructure capabilities; the
26 balance will be similar to that for existing management methods.
- 27 b) The cost factors will be similar to those of existing management methods.
- 28 c) This project adds YANG capabilities to IEEE Std 802.1AX as a step towards a complete YANG
29 management solution. This helps to eliminate multiple management platforms, thus reduces
30 installation cost.
- 31 d) This project adds YANG capabilities to IEEE Std 802.1AX as a step towards a complete YANG
32 management solution. This helps to eliminate multiple management platforms, thus reduces
33 operational cost.

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4 working group meetings. Preparation of drafts often exposes inconsistencies in editor's instructions or
5 exposes the need to make choices between approaches that were not fully apparent in the meeting. Choices
6 and requests by the editors' for contributions on specific issues will be found in the editors' [Introduction to the](#)
7 [current draft](#) and at appropriate points in the draft.

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9 working group drafts, are part of the audit trail of the development of the standard and are available, along
10 with all the revisions of the draft on the 802.1 website (for address see above).

11 Records of participants in the development of the standard are added after SA Ballot, as part of
12 pre-publication editing by IEEE Staff.

13 MIB and YANG modules

14 The MIB and YANG modules that are modified or added by this amendment are attached to the draft pdf as
15 plain text (UTF-8) .mib and .yang files. When a roll up of the current base standard plus this amendment is
16 made available, all the MIB and YANG modules for the roll up are attached.

1 Introduction to the current draft¹

2 This introduction is not part of the draft, and should not be the subject of ballot comments.

3 D1.1

4 Incorporates the resolution of comments received during the Working Group ballot of draft 1.0. Significant
5 changes resulting from the ballot include:

- 6 — Changing the representation of the 4096 bit mask in the drni-mask grouping from binary to a string
7 listing individual CIDs, or ranges of CIDs, corresponding to bits in the mask that are set to 1.
- 8 — Removing several 4096 bit masks from the YANG model because they could be calculated from other
9 information in the model;
- 10 — Moving the actor-admin-state leaf from the aggpport container to the key-group container.

11

12 D1.0

13 Incorporates the resolution of comments received during the Task Group ballot of draft 0.2. Significant
14 changes resulting from the ballot include:

- 15 — Combining the old figures 10-2 and 10-3 into a single page UML-like diagram for Link Aggregation
16 (excluding DRNI);
- 17 — Clean up separation of Link Aggregation from the optional DRNI by adding a separate line in the
18 conformance clause options for DRNI, and moving all DRNI related types from
19 *ieee802-dot1ax-types.yang* to *ieee802-dot1ax-drni.yang*;
- 20 — Moving the system and system-priority leafs from key-groups to a separate list of aggregation
21 systems. This allows the system mac address to be provided by the system rather than requiring it to
22 be explicitly specified in a configuration model.

23 Also changed was adding a base identity choice for the admin-conv-link-map and admin-conv-service-map of
24 CSCD. This allows a vendor to augment the yang model with an identity statement specifying pre-defined
25 contents of the table to provide a shorthand for configuration that otherwise could potentially take thousands
26 of lines of configuration.

27 D0.2

28 Incorporates the resolution of comments received during the Task Group ballot of draft 0.1.

29 D0.1

30 This is an initial draft and comments are requested on all aspects of the draft. It includes a number of notes
31 that may be of help to the Editors as well as informing the initial review process.

32 Stephen Haddock, 802.1AXdz Editor

33

¹ The whole or parts of the introduction, possibly updated, to past drafts may be retained at the Editor's discretion, with the most recent introduction first. The introduction to each draft may solicit input on specific subjects.

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7 **Local and metropolitan area networks—**

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1

2 **Abstract:** This amendment to IEEE Std 802.1AX-2020 specifies a Unified Modeling Language
3 (UML)-based model and YANG modules for Link Aggregation configuration and status reporting.

4 **Keywords:** Aggregated Link, Aggregator, Distributed Resilient Network Interconnect, DRNI,
5 interconnect, Link Aggregation, Link Aggregation Group, local area network, management,
6 Network-Network Interface, NNI, YANG.

7

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3 At the time this standard was submitted to the IEEE-SA Standards Board for approval, the IEEE 802.1
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9 **Stephen Haddock**, *Editor*
10

<<TBA>>

¹ The following members of the individual balloting committee voted on this standard. Balloters may have
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⁴ membership:

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⁸
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¹⁰

1 Introduction

This introduction is not part of IEEE Std 802.1AXdz™-20XX, IEEE Standard for Local and metropolitan area networks—Link Aggregation—Amendment 1: YANG for Link Aggregation.

2 IEEE Std 802.1AXdz™-202X: YANG for Link Aggregation specifies a Unified Modeling Language
3 (UML)-based model and YANG modules for Link Aggregation configuration and status reporting

4 This standard contains state-of-the-art material. The area covered by this standard is undergoing evolution.
5 Revisions are anticipated within the next few years to clarify existing material, to correct possible errors, and
6 to incorporate new related material. Information on the current revision state of this and other IEEE 802
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------------	-------------------------------------	----

¹ 2. Normative references

² *Insert the following references into Clause 2 in alphanumeric order:*

³ IETF RFC 6241, Network Configuration Protocol (NETCONF), June 2011.

⁴ IETF RFC 6991, Common YANG Data Types, July 2013

⁵ IETF RFC 7950, The YANG 1.1 Data Modeling Language, August 2016.

⁶ IETF RFC 8343, A YANG Data Model for Interface Management, March 2018

⁷ IETF RFC 8349, A YANG Data Model for Routing Management (NMDA Version), March 2018

¹ 4. Acronyms and abbreviations

² *Insert the following acronyms into Clause 4 in alphabetical order:*

³ NETCONF Network Configuration Protocol

⁴ UML® Unified Modeling Language™¹

¹ UML® is a registered trademark of the Object Management Group, Inc., and Unified Modeling Language™ is a trademark of the Object Management Group, Inc.

5. Conformance

Insert the following text (items j and k) after item i) in the lettered list of 5.3.2

5.3.2 Link Aggregation options

- j) Support YANG modules for the management of Link Aggregation capabilities (Clause 10).
- k) Support YANG modules for the management of DRNI capabilities (Clause 10).

1 *Insert Clause 10 after Clause 9 as follows:*

2 **10. YANG Data Model**

3 This clause specifies the YANG data model comprised of YANG modules that provide control and status
4 monitoring of systems and system components that implement functionality specified in this standard.

5 This clause:

- 6 a) Introduces the YANG framework that governs the naming and hierarchy of configuration and
7 operational data structures in the data models, and the modeling of network interfaces (10.1).
- 8 b) Describes the information data model and its relationship to the operational processes and managed
9 objects specified in the other clauses of this standard, and provides a representation, similar to the
10 Unified Modeling Language (UML), of relevant objects (10.2).
- 11 c) Describes the structure of the YANG data model, which comprises three YANG modules (10.3).
- 12 d) Includes a relationship description of other modules imported in YANG modules (10.4)
- 13 e) Reviews security considerations applicable to each of the modules, with specific reference to data
14 nodes in the YANG modules that compose the YANG data model (10.5).
- 15 f) Includes each of the YANG modules and its data schema (10.6).

16 **10.1 YANG Framework**

17 The YANG framework applies hierarchy in the following areas:

- 18 a) The uniform resource name (URN), as specified in 802d.
- 19 b) The YANG objects form a hierarchy of configuration and operational data structures that specify the
20 YANG model.

21 **10.2 Information Model for Link Aggregation Management**

22 The YANG objects are based on the managed objects in Clause 7. A UML-like representation of the
23 management model is provided in the following subclauses.

24 NOTE—OMG® UML2.5² [B11] conventions together with C++ language constructs are used in this clause as a
25 representation to convey model structure and relationships..

26 The purpose of an UML-like³ diagram is to express the model design on a single piece of paper. The
27 structure of the UML-like representation shows the name of the object followed by a list of properties for the
28 object. The properties indicate its type and accessibility. The UML-like representation is meant to express
29 simplified semantics for the properties. It is not meant to provide the specific datatype as used to encode the
30 object in either MIB or YANG. In the UML-like representation, a box with a white background represents
31 information that comes from sources outside of the IEEE. A box with a gray background represents objects
32 that are specified by this IEEE Standard.

33 The YANG hierarchical structure that incorporates the Link Aggregation YANG modules supported by this
34 standard is represented by Figure 10-1. In the figures in this clause, items that are shaded gray are described
35 in this document, items with no background shading are specified elsewhere. The YANG data model is

²OMG® is a registered trademark of the Object Management Group, Inc.

³A description of the UML-like diagrams used in this clause is provided at <https://1.ieee802.org/uml-like-diagrams>.

1 realized in three YANG modules. One module *ieee802-dot1ax-types* provides data types that are needed by
2 the Link Aggregation configuration and monitoring objects. The *ieee802-dot1ax-linkagg* module provides
3 the Link Aggregation, Aggregation Port, and Aggregator configuration and monitoring objects. The
4 *ieee802-dot1ax-drni* module provides the Distributed Resilient Network Interface (DRNI) configuration and
5 monitoring objects. The Link Aggregation and DRNI capabilities are not only applicable to IEEE Std
6 802.1Q bridges, but also (for example) end stations, and routers.

7

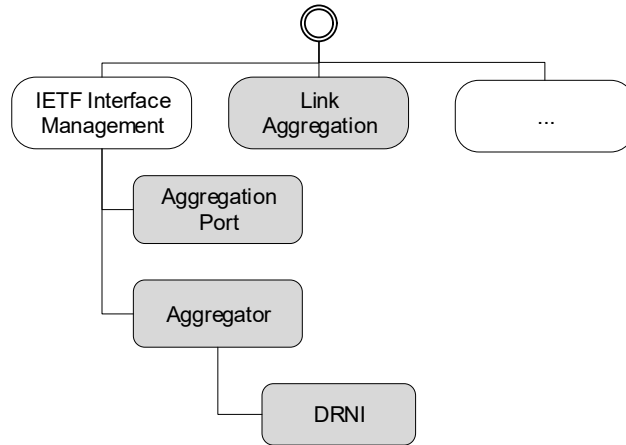


Figure 10-1—YANG root hierarchy with Link Aggregation YANG modules

8 10.2.1 Link Aggregation model

9 The Link Aggregation Configuration and Monitoring Objects in Figure 10-2 show the objects that are
10 applicable on a system supporting link aggregation, and the interfaces supporting link aggregation.

11 The objects on the system consist of a list of aggregation systems, and a list of key groups. A key group
12 is the set of aggregators and aggregation ports that share the same system priority, system mac address, and
13 aggregation key, and therefore can potentially form a Link Aggregation Group. Each key group has a unique
14 combination of actor-admin-key and actor-system, and includes parameters that have the same value for any
15 aggregator and/or aggregation port that have the same actor-admin-key and actor-system.

16 NOTE—See 6.3.5 and 6.3.6 for a description of the identification of Aggregation Ports that can potentially form a Link
17 Aggregation Group, and Figure 6-15 for examples.

18 Each Aggregation Port is an interface (e.g. of type ethernetCsmacd) augmented with an “aggport” presence
19 container as shown on the right side of Figure 10-2. Each Aggregator is an interface to the entire Link
20 Aggregation Group (typically using type ieee8023adlag) augmented with a “lag” container as shown on the
21 left side of Figure 10-2. The binding of Aggregation Ports to Aggregators is a dynamic function of the Link
22 Aggregation Control Protocol. The Aggregation Ports attached to an Aggregator can be read from the *lower-*
23 *layer-if* attribute of the Aggregator interface. The Aggregator to which an Aggregation Port is attached can
24 be read through the *higher-layer-if* attribute of the Aggregation Port interface.

25

1

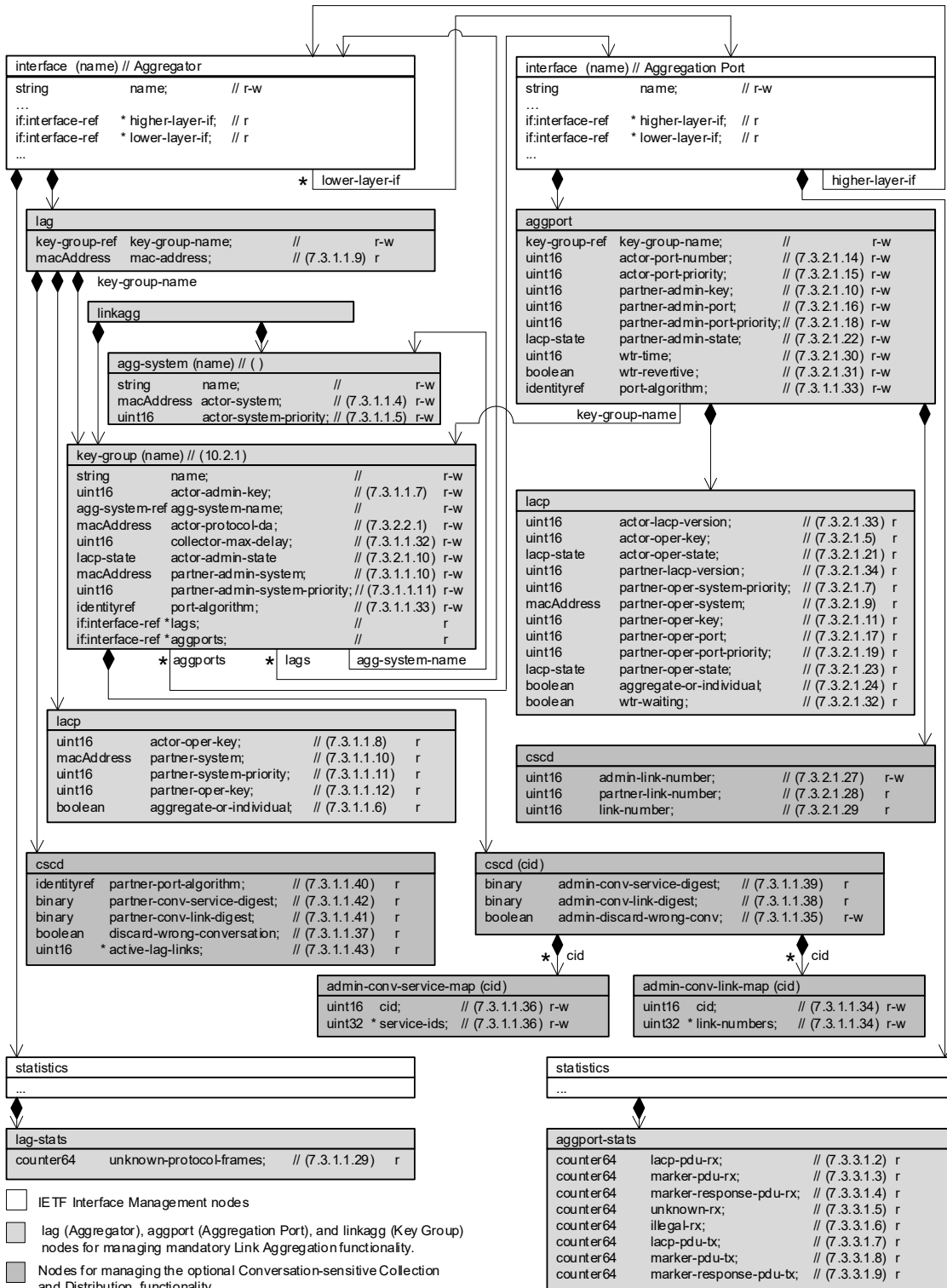


Figure 10-2—Link Aggregation Configuration and Monitoring Objects

10.2.2 DRNI model

The DRNI Configuration and Monitoring Objects in Figure 10-3 show the objects that are applicable to an Aggregator augmented with a “drni” container.

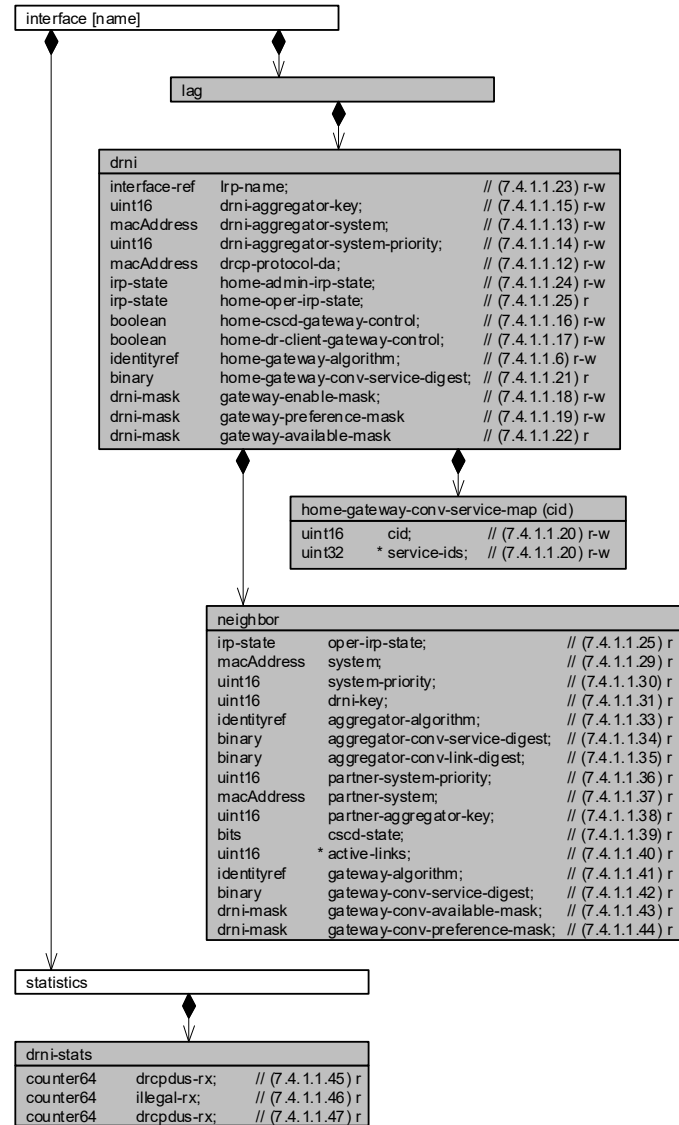


Figure 10-3—DRNI Configuration and Monitoring Objects

10.3 Structure of the Link Aggregation YANG Model

The YANG data model specified in this standard is comprised of three YANG modules. A summary of the modules contained in this clause is represented in Table 10-1.

Table 10-1—Structure of the YANG modules

Module	Subclause	Notes
ieee802-dot1ax-types	10.7.1	Type definitions used for Link Aggregation YANG.
ieee802-dot1ax-linkagg	10.7.2	Link Aggregation Management
ieee802-dot1ax-drni	10.7.3	DRNI Management

In the YANG modules below, if any discrepancy between the DESCRIPTION text and the corresponding specification in any other part of this standard occurs, the specifications outside this clause take precedence.

10.4 Relationship to other YANG modules

This clause describes how the *ieee802-dot1ax-linkagg* and *ieee802-dot1ax-drni* YANG modules are related to the YANG modules that are imported.

10.4.1 IEEE 802.1AX Types Module

The *ieee802-dot1ax-types* module provides reusable types that are used by the *ieee802-dot1ax-linkagg* and *ieee802-dot1ax-drni* modules.

10.4.2 IETF YANG Types Module

The *ietf-yang-types* YANG module (IETF RFC 6991) contains a set of derived YANG types. This document leverages counter64.

10.4.3 IETF Interfaces YANG Module

The *ietf-interfaces* YANG module (IETF RFC 8343) contains a set of YANG definitions for managing network interfaces. This document augments an *ietf-interfaces:interface* with aggregation-port or aggregator data nodes..

10.4.4 IEEE 802 Types Module

The *ieee802-types* module provides reusable types that are used in IEEE 802 standards.

NOTE—The type for mac-addresses defined in *ieee802-types* has a pattern that allows upper and lower case letters. To avoid issues with string comparison, it is suggested to only use upper case for the letters in the hexadecimal numbers. Implementers using code comparing MAC addresses are advised that there is still an issue with a difference between the IETF mac-address definition and the IEEE mac-address definition.

10.5 Security Considerations

The YANG modules specified in this clause are designed to be accessed via a network configuration protocol (e.g., NETCONF protocol). In the case of NETCONF, the lowest NETCONF layer is the secure transport layer and the mandatory to implement secure transport is SSH. The NETCONF access control model provides the means to restrict access for particular NETCONF users to a pre-configured subset of all available NETCONF protocol operations and content.

It is the responsibility of a system's implementor and administrator to ensure that the protocol entities in the system that support NETCONF, and any other remote configuration protocols that make use of these YANG modules, are properly configured to allow access only to those users who have legitimate rights to read or write data nodes. This standard does not specify how the credentials of those users are to be stored or validated.

10.5.1 Security considerations of the *ieee802-dot1ax* YANG modules

There are several management objects specified in the *ieee802-dot1ax-linkagg* and *ieee802-dot1ax-drni* YANG modules that are configurable (i.e., read-write) and/or operational (i.e., read-only). Such objects could be considered sensitive or vulnerable in some network environments. A network configuration protocol, such as NETCONF (IETF RFC 6241), can support protocol operations that can edit or delete YANG module configuration data (e.g., edit-config, delete-config, copy-config). If this is done in a non-secure environment without proper protection, then negative effects on the network operation are possible.

The following containers, and the objects in these containers, of the *ieee802-dot1ax-linkagg* and *ieee802-dot1ax-drni* YANG modules can be manipulated to interfere with the operation of the Link Aggregation Control Protocol (LACP) or the Distributed Relay Control Protocol (DRCP). This could, for example, cause network instability and result in the loss of service for a large number of end users.

- `dot1ax-linkagg/link-aggregation/key-group`
- `dot1ax-linkagg/link-aggregation/cscd`
- `dot1ax-linkagg/link-aggregation/cscd/admin-conv-service-map`
- `dot1ax-linkagg/link-aggregation/cscd/admin-conv-link-map`
- `dot1ax-linkagg/aggregation-port`
- `dot1ax-linkagg/aggregation-port/cscd/admin-link-number`
- `dot1ax-linkagg/aggregator`
- `dot1ax-drni/aggregator/drni`
- `dot1ax-drni/aggregator/drni/home-admin-conv-service-map`

Some of the readable data in this YANG module could be considered sensitive or vulnerable in some network environments. It is important to control all types of access (e.g., including NETCONF get, get-config operations) to these objects and possibly to even encrypt the values of these objects when sending them over the network. For example, the system name and other information about the remote systems could provide information about the configuration and topology of the network and could be considered a privacy threat.

10.6 YANG schema tree definitions^{4,5}

A simplified graphical representation of the data model is used in this document. The meaning of the symbols in these diagrams is as follows:

- Brackets "[" and "]" enclose list keys.
- Abbreviations before data node names: "rw" means configuration (read-write), and "ro" means state data (read-only).
- Symbols after data node names: "?" means an optional node, "!" means a presence container, and "*" denotes a list and leaf-list.
- Parentheses enclose choice and case nodes, and case nodes are also marked with a colon (":").
- Ellipsis ("...") stand for contents of subtrees that are not shown.

10.6.1 Schema for the *ieee802-dot1ax-linkagg* YANG module

```
module: ieee802-dot1ax-linkagg
  +--rw linkagg
    +--rw agg-system* [name]
      | +--rw name string
      | +--rw actor-system? ieee:mac-address
      | +--rw actor-system-priority? uint16
    +--rw key-group* [name]
      +--rw name string
      +--rw actor-admin-key uint16
      +--rw agg-system-name agg-system-ref
      +--rw actor-protocol-da? ieee:mac-address
      +--rw collector-max-delay? uint16
      +--rw actor-admin-state? ax:lacp-state
      +--rw partner-admin-system? ieee:mac-address
      +--rw partner-admin-system-priority? uint16
      +--rw port-algorithm? identityref
      +--ro lags* if:interface-ref
      +--ro aggports* if:interface-ref
      +--rw cscd {ax:cscd}?
        +--rw admin-conv-service-map {ax:sid-map}?
          | +--rw (method)?
          | | +--rw pattern? identityref
          | | +--rw (cid-list)
          | | +--rw cid-list* [cid]
          | | +--rw cid uint16
          | | +--rw service-ids* uint32
          +--ro admin-conv-service-digest? binary
        +--rw admin-conv-link-map
          | +--rw (method)?
          | | +--rw pattern? identityref
          | | +--rw (cid-list)
          | | +--rw cid-list* [cid]
          | | +--rw cid uint16
          | | +--rw link-numbers* uint16
          +--ro admin-conv-link-digest? binary
        +--rw admin-discard-wrong-conv? enumeration {ax:dwc}?
    augment /if:interfaces/if:interface:
      +--rw lag
        +--rw key-group-name key-group-ref
        +--ro mac-address? ieee:mac-address
        +--ro lacp
        | +--ro actor-oper-key? uint16
```

⁴Copyright release for YANG modules: Users of this standard may freely reproduce the YANG modules contained in this subclause so that they can be used for their intended purpose.

⁵An ASCII version of the YANG module(s) can be obtained by Web browser from the IEEE 802.1 Website at <https://1.ieee802.org/yang-modules/>.

```

1      | +--ro partner-system?          ieee:mac-address
2      | +--ro partner-system-priority? uint16
3      | +--ro partner-oper-key?       uint16
4      | +--ro aggregate-or-individual? boolean
5      +--ro cscd {ax:cscd}?
6          +--ro partner-port-algorithm? identityref
7          +--ro partner-conv-service-digest? binary
8          +--ro partner-conv-link-digest? binary
9          +--ro discard-wrong-conversation? boolean
10         +--ro active-lag-links*      uint16
11 augment /if:interfaces/if:interface/if:statistics:
12     +--ro lag-stats
13         +--ro unknown-protocol-frames? yang:counter64
14 augment /if:interfaces/if:interface:
15     +--rw aggport!
16         +--rw key-group-name          key-group-ref
17         +--rw actor-port-number?      uint16
18         +--rw actor-port-priority?    uint16
19         +--rw partner-admin-key?      uint16
20         +--rw partner-admin-port?     uint16
21         +--rw partner-admin-port-priority? uint16
22         +--rw partner-admin-state?    ax:lacp-state
23         +--rw wtr-time?               uint16
24         +--rw wtr-revertive?          boolean
25         +--ro lacp
26         | +--ro actor-lacp-version?   uint8
27         | +--ro actor-oper-key?       uint16
28         | +--ro actor-oper-state?     ax:lacp-state
29         | +--ro partner-lacp-version? uint8
30         | +--ro partner-oper-system-priority? uint16
31         | +--ro partner-oper-system?  ieee:mac-address
32         | +--ro partner-oper-key?     uint16
33         | +--ro partner-oper-port?    uint16
34         | +--ro partner-oper-port-priority? uint16
35         | +--ro partner-oper-state?   ax:lacp-state
36         | +--ro aggregate-or-individual? boolean
37         | +--ro wtr-waiting?          boolean
38         +--rw cscd {ax:cscd}?
39             +--rw admin-link-number?  uint16
40             +--ro partner-link-number? uint16
41             +--ro link-number?        uint16
42 augment /if:interfaces/if:interface/if:statistics:
43     +--ro aggport-stats
44         +--ro lacp-pdu-rx?            yang:counter64
45         +--ro marker-pdu-rx?         yang:counter64
46         +--ro marker-response-pdu-rx? yang:counter64
47         +--ro unknown-rx?            yang:counter64
48         +--ro illegal-rx?            yang:counter64
49         +--ro lacp-pdu-tx?           yang:counter64
50         +--ro marker-pdu-tx?         yang:counter64
51         +--ro marker-response-pdu-tx? yang:counter64
52

```

53 10.6.2 Schema for the *ieee802-dot1ax-drni* YANG module

```

54 module: ieee802-dot1ax-drni
55
56 augment /if:interfaces/if:interface/dot1ax:lag:
57     +--rw drni!
58         +--rw irp-name                if:interface-ref
59         +--rw drni-aggregator-key?    uint16
60         +--rw drni-aggregator-system? ieee:mac-address
61         +--rw drni-aggregator-system-priority? uint16
62         +--rw drcp-protocol-da?       ieee:mac-address
63         +--rw home-admin-irp-state?   irp-state
64         +--ro home-oper-irp-state?    irp-state
65         +--rw home-cscd-gateway-control? boolean
66         +--rw home-dr-client-gateway-control? boolean
67         +--rw home-gateway-algorithm? identityref
68         +--rw home-gateway-conv-service-map {ax:sid-map}?
69         | +--rw (method)?

```

```

1      |      +---:(pattern)
2      |      | +--rw pattern?      identityref
3      |      +---:(cid-list)
4      |      | +--rw cid-list* [cid]
5      |      | +--rw cid          uint16
6      |      | +--rw service-ids* uint32
7      +---ro home-gateway-conv-service-digest?  binary
8      +---rw gateway-enable-mask
9      | +--rw cid-str*      string
10     | +--rw invert-mask?   boolean
11     +---rw gateway-preference-mask
12     | +--rw cid-str*      string
13     | +--rw invert-mask?   boolean
14     +---ro gateway-available-mask
15     | +--ro cid-str*      string
16     | +--ro invert-mask?   boolean
17     +---ro neighbor
18         +---ro oper-irp-state?      irp-state
19         +---ro system?               ieee:mac-address
20         +---ro system-priority?      uint16
21         +---ro drni-key?             uint16
22         +---ro aggregator-algorithm? identityref
23         +---ro aggregator-conv-service-digest? binary
24         +---ro aggregator-conv-link-digest?  binary
25         +---ro partner-system-priority? uint16
26         +---ro partner-system?         ieee:mac-address
27         +---ro partner-aggregator-key?   uint16
28         +---ro cscd-state?              bits
29         +---ro active-links*            uint16
30         +---ro gateway-algorithm?       identityref
31         +---ro gateway-conv-service-digest? binary
32         +---ro gateway-available-mask
33         | +--ro cid-str*      string
34         | +--ro invert-mask?   boolean
35         +---ro gateway-preference-mask
36         | +--ro cid-str*      string
37         | +--ro invert-mask?   boolean
38     augment /if:interfaces/if:interface/if:statistics:
39         +---ro drni-stats
40         | +--ro drcpdus-rx?   yang:counter64
41         | +--ro illegal-rx?   yang:counter64
42         | +--ro drcpdus-tx?   yang:counter64
43

```

44 10.7 YANG modules

45 10.7.1 The *ieee802-dot1ax-types* YANG module

```

46 module ieee802-dot1ax-types {
47   yang-version 1.1;
48   namespace "urn:ieee:params:xml:ns:yang:ieee802-dot1ax-types";
49   prefix dot1ax-types;
50
51   organization
52     "IEEE 802.1 Working Group";
53   contact
54     "WG-URL: http://www.ieee802.org/1/
55      WG-Email: stds-802-1-1@ieee.org
56
57     Contact: IEEE 802.1 Working Group Chair
58     Postal: C/O IEEE 802.1 Working Group
59            IEEE Standards Association
60            45 Hoes Lane
61            Piscataway, NJ 08854
62            USA
63     E-mail: stds-802-1-chairs@ieee.org";
64   description
65     "Common types used within 802.1AX Link Aggregation modules.
66

```

```
1      Copyright (C) IEEE (2024).
2
3      This version of this YANG module is part of IEEE Std 802.1AX;
4      see the standard itself for full legal notices.";
5
6      revision 2025-02-07 {
7          description
8              "Published as part of IEEE Std 802.1AXdz-202y.
9              The following reference statement identifies each referenced IEEE
10             Standard as updated by applicable amendments.";
11          reference
12              "IEEE Std 802.1AX-2020 Link Aggregation:
13              IEEE Std 802.1AXdz-202y.";
14      }
15
16      feature csdc {
17          description
18              "Conversation Sensitive Collection and Distribution (CSCD)
19              is supported.";
20          reference
21              "5.3.2, 6.6 of IEEE Std 802.1AX";
22      }
23
24      feature dwc {
25          description
26              "The Discard Wrong Conversation option in CSCD is
27              supported.";
28          reference
29              "5.3.2, 6.6 of IEEE Std 802.1AX";
30      }
31
32      feature sid-map {
33          description
34              "The Service ID to Conversation ID map is supported for
35              use by distribution algorithms.";
36      }
37
38      feature c-vid-dist-alg {
39          description
40              "Supports distribution based on C-VIDs.";
41      }
42
43      feature s-vid-dist-alg {
44          description
45              "Supports distribution based on S-VIDs.";
46      }
47
48      feature i-sid-dist-alg {
49          description
50              "Supports distribution based on I-SIDs.";
51      }
52
53      feature te-sid-dist-alg {
54          description
55              "Supports distribution based on TE-SIDs.";
56      }
57
58      feature flow-hash-dist-alg {
59          description
60              "Supports distribution based on Flow Hash.";
61      }
62
63      feature basic-link-maps {
64          description
65              "Pre-defined admin-conv-link-map configurations to distribute
66              packets on either one or two active links, with any
67              remaining links available as standby.";
68      }
69
70      identity distribution-algorithm {
71          description
72              "Each distribution algorithm is identified by a sequence of
```

```
1      4 octets, structured as shown in Figure 8-1. Distribution
2      algorithm identifiers are used by network administrators to
3      select between algorithms and, in Conversation-sensitive
4      LACP and (if supported) Distributed Resilient Network
5      Interconnect (DRNI) operation, to check whether partners
6      and neighbors are using the same algorithm.
7
8      This identity is intended to serve as base identity, not
9      to be directly referenced.
10
11     Vendor specific, combination (ex: multi-layer), and other
12     customized distribution algorithms can be created as
13     their own identities in their own YANG files, derived from
14     this imported base type. When the identity description
15     specifies a 4 octet value, this value is used for cscd
16     purposes, and included in LACPDUs. Otherwise the identity
17     is used purely for local selection of a distribution
18     algorithm, and the unspecified distribution algorithm is
19     used for cscd purposes.";
20     reference
21       "8.1, 8.2 of IEEE Std 802.1AX";
22   }
23
24   identity unspecified {
25     base distribution-algorithm;
26     description
27       "The 'Unspecified distribution algorithm' identifier has
28       been reserved for use when the algorithm is unknown (or
29       is not advertised).
30       The algorithm identifier is 00-80-C2-00.";
31     reference
32       "Table 8-1 of IEEE Std 802.1AX";
33   }
34
35   identity c-vids-nomap {
36     if-feature "c-vid-dist-alg";
37     base distribution-algorithm;
38     description
39       "Distribution based on C-VIDs (8.2.1). No Service ID
40       mapping table is used.
41       The algorithm identifier is 00-80-C2-01.";
42     reference
43       "Table 8-1 of IEEE Std 802.1AX";
44   }
45
46   identity c-vids-map {
47     if-feature "c-vid-dist-alg and sid-map";
48     base distribution-algorithm;
49     description
50       "Distribution based on C-VIDs (8.2.1). A Service ID
51       mapping table is used.
52       The algorithm identifier is 00-80-C2-81.";
53     reference
54       "Table 8-1 of IEEE Std 802.1AX";
55   }
56
57   identity s-vids-nomap {
58     if-feature "s-vid-dist-alg";
59     base distribution-algorithm;
60     description
61       "Distribution based on S-VIDs (8.2.2). No Service ID
62       mapping table is used.
63       The algorithm identifier is 00-80-C2-02.";
64     reference
65       "Table 8-1 of IEEE Std 802.1AX";
66   }
67
68   identity s-vids-map {
69     if-feature "s-vid-dist-alg and sid-map";
70     base distribution-algorithm;
71     description
72       "Distribution based on S-VIDs (8.2.2). A Service ID
```

```
1      mapping table is used.
2      The algorithm identifier is 00-80-C2-82.";
3  reference
4      "Table 8-1 of IEEE Std 802.1AX";
5  }
6
7  identity i-sids-nomap {
8      if-feature "i-sid-dist-alg";
9      base distribution-algorithm;
10     description
11         "Distribution based on I-SIDs (8.2.3). No Service ID
12         mapping table is used.
13         The algorithm identifier is 00-80-C2-03.";
14     reference
15         "Table 8-1 of IEEE Std 802.1AX";
16 }
17
18 identity i-sids-map {
19     if-feature "i-sid-dist-alg and sid-map";
20     base distribution-algorithm;
21     description
22         "Distribution based on I-SIDs (8.2.3). A Service ID
23         mapping table is used.
24         The algorithm identifier is 00-80-C2-83.";
25     reference
26         "Table 8-1 of IEEE Std 802.1AX";
27 }
28
29 identity te-sids-nomap {
30     if-feature "te-sid-dist-alg";
31     base distribution-algorithm;
32     description
33         "Distribution based on TE-SIDs (8.2.4). No Service ID
34         mapping table is used.
35         The algorithm identifier is 00-80-C2-04.";
36     reference
37         "Table 8-1 of IEEE Std 802.1AX";
38 }
39
40 identity te-sids-map {
41     if-feature "te-sid-dist-alg and sid-map";
42     base distribution-algorithm;
43     description
44         "Distribution based on TE-SIDs (8.2.4). A Service ID
45         mapping table is used.
46         The algorithm identifier is 00-80-C2-84.";
47     reference
48         "Table 8-1 of IEEE Std 802.1AX";
49 }
50
51 identity flow-hash-nomap {
52     if-feature "flow-hash-dist-alg";
53     base distribution-algorithm;
54     description
55         "Distribution based on Flow Hash (8.2.5). No Service ID
56         mapping table is used.
57         The algorithm identifier is 00-80-C2-05.";
58     reference
59         "Table 8-1 of IEEE Std 802.1AX";
60 }
61
62 identity flow-hash-map {
63     if-feature "flow-hash-dist-alg and sid-map";
64     base distribution-algorithm;
65     description
66         "Distribution based on Flow Hash (8.2.5). A Service ID
67         mapping table is used.
68         The algorithm identifier is 00-80-C2-85.";
69     reference
70         "Table 8-1 of IEEE Std 802.1AX";
71 }
72
```



```
1 identity link-map-patterns {
2   description
3     "Base identity for patterns filling the admin-conv-link-map.
4     This identity is intended to serve as base identity, not
5     to be directly referenced. Use of the identity allows vendors
6     to augment the module with vendor specified patterns,";
7 }
8
9 identity one-plus-n {
10  if-feature "basic-link-maps";
11  base link-map-patterns;
12  description
13    "Provides active/standby behavior with one active link and
14    up to 63 standby links. All packets are mapped to whichever
15    link that is active in the aggregation has the lowest link
16    number. If that link fails, all packets are mapped to the
17    lowest link number of the remaining links that are active
18    in the aggregation.
19    The map consists of an identical entry for each CID, with a
20    list of numbers from 1 to 64.
21    Supports any number of links in the aggregation (up to 64)
22    with link numbers between 1 and 64. Any links in the
23    aggregation with a link number greater than 64 are not
24    used.";
25 }
26
27 identity two-plus-n {
28  if-feature "basic-link-maps";
29  base link-map-patterns;
30  description
31    "Provides basic load-balancing on two active links with up
32    to 62 standby links. All packets with an even CID value are
33    mapped to the link with the lowest link number in the
34    aggregation, and all packets with an odd CID value are
35    mapped to the link with the highest link number in the
36    aggregation.
37    The map consists of an identical entry for each even CID
38    containing a list of numbers from 1 to 64, and an identical
39    entry for each odd CID containing a list of numbers from 64
40    to 1.
41    Supports any number of links in the aggregation (up to 64)
42    with link numbers between 1 and 64. Any links in the
43    aggregation with a link number greater than 64 are not
44    used.";
45 }
46
47 identity service-map-patterns {
48  description
49    "Base identity for patterns filling the admin-conv-service-map.
50    This identity is intended to serve as base identity, not
51    to be directly referenced. Use of the identity allows vendors
52    to augment the module with vendor specified patterns,";
53 }
54
55 typedef lacp-state {
56  type bits {
57    bit lacp-activity {
58      position 1;
59      description
60        "Provides administrative control over when LACPDUs are
61        transmitted. A value of '1' indicates Active mode where
62        LACPDUs are sent regardless of partner's lacp-activity
63        value. A value of '0' indicates Passive mode where
64        LACPDUs are sent only when the partner's lacp-activity
65        value is '1' (partner is in Active mode).";
66    }
67    bit lacp-timeout {
68      position 2;
69      description
70        "Provides administrative control over the frequency of
71        received LACPDUs. A value of '1' indicates Short Timeout
72        (so partner uses frequent transmission). A value of '0'
```

```

1         indicates Long Timeout (so partner can use infrequent
2         transmission).";
3     }
4     bit aggregation {
5         position 3;
6         description
7             "Provides administrative control over whether this
8             Aggregation Port can be in a LAG with more than one
9             member. A value of '1' indicates the port can be
10            aggregated with other ports. A value of '0' indicates
11            the port can only be a solitary link.";
12    }
13    bit synchronization {
14        position 4;
15        description
16            "The Synchronization state of the MUX state machine.";
17    }
18    bit collecting {
19        position 5;
20        description
21            "The Collecting state of the MUX state machine.";
22    }
23    bit distributing {
24        position 6;
25        description
26            "The Distributing state of the MUX state machine.";
27    }
28    bit defaulted {
29        position 7;
30        description
31            "Indicates the port is using the partner-admin values
32            to select an Aggregator.";
33    }
34    bit expired {
35        position 8;
36        description
37            "The Expired state of the Receive state machine.";
38    }
39 }
40 description
41     "LACP state values as transmitted in LACPDUs.
42     Administrative control over the values of lacp-activity,
43     lacp-timeout, aggregation, and (in partner-admin-lacp-state)
44     synchronization is allowed.
45     The remaining bits are read-only.";
46 reference
47     "6.4.1, 6.4.2.3 of IEEE Std 802.1AX";
48 }
49
50 grouping link-map {
51     description
52         "Specifies the contents of the admin-conv-link-map.";
53     choice method {
54         default "cid-list";
55         description
56             "Provides two ways to specify the map contents.";
57         leaf pattern {
58             type identityref {
59                 base link-map-patterns;
60             }
61             description
62                 "Use a predefined pattern to fill the map.";
63         }
64         list cid-list {
65             key "cid";
66             description
67                 "Data structure to map a Conversation Identifier
68                 (CID) to a Link Number. Each entry consists of a CID
69                 and a list of link numbers that can potentially be
70                 selected for that CID. An empty list of link-numbers
71                 means that no links are selected for the CID.";
72             leaf cid {

```

```

1      type uint16 {
2          range "0..4095";
3      }
4      description
5          "Port Conversation Identifier";
6  }
7  leaf-list link-numbers {
8      type uint16;
9      description
10         "List of zero or more Link Numbers that can
11         potentially be selected for distribution of frames
12         with this CID.";
13  }
14  }
15  }
16  }
17
18  grouping service-map {
19      description
20          "Specifies the contents of the admin-conv-service-map.";
21      choice method {
22          default "cid-list";
23          description
24              "Provides two ways to specify the map contents.";
25          leaf pattern {
26              type identityref {
27                  base service-map-patterns;
28              }
29              description
30                  "Use a predefined pattern to fill the map.";
31          }
32          list cid-list {
33              key "cid";
34              description
35                  "Data structure to map service identifiers to
36                  conversation identifiers. Each entry consists of a
37                  Conversation ID (CID) and a list of zero or more
38                  Service Identifiers (SIDs) that map to it. An empty
39                  list of SIDs means there are no SIDs that map to this
40                  CID, and results in the same behavior as not having an
41                  entry for this CID.";
42              leaf cid {
43                  type uint16 {
44                      range "0..4095";
45                  }
46                  description
47                      "Port Conversation Identifier";
48              }
49              leaf-list service-ids {
50                  type uint32;
51                  description
52                      "List of zero or more SIDs that map to the CID.";
53              }
54          }
55      }
56  }
57 }
58
59

```

60 10.7.2 The *ieee802-dot1ax-linkagg* YANG module

```

61 module ieee802-dot1ax-linkagg {
62     yang-version 1.1;
63     namespace "urn:ieee:params:xml:ns:yang:ieee802-dot1ax-linkagg";
64     prefix dot1ax;
65
66     import ieee802-dot1ax-types {
67         prefix ax;
68     }
69     import ieee802-types {

```

```
1  prefix ieee;
2  }
3  import ietf-yang-types {
4    prefix yang;
5  }
6  import ietf-interfaces {
7    prefix if;
8  }
9  import iana-if-type {
10   prefix ianaif;
11 }
12
13 organization
14   "IEEE 802.1 Working Group";
15 contact
16   "WG-URL: http://www.ieee802.org/1/
17   WG-Email: stds-802-1-1@ieee.org
18
19   Contact: IEEE 802.1 Working Group Chair
20   Postal: C/O IEEE 802.1 Working Group
21           IEEE Standards Association
22           45 Hoes Lane
23           Piscataway, NJ 08854
24           USA
25   E-mail: stds-802-1-chairs@ieee.org";
26 description
27   "This YANG module describes the configuration model for Link
28   Aggregation, as specified in IEEE Std 802.1AX, including Link
29   Aggregation Control Protocol (LACP) and Conversation Sensitive
30   Collection and Distribution.
31
32   Copyright (C) IEEE (2024).
33
34   This version of this YANG module is part of IEEE Std 802.1AX;
35   see the standard itself for full legal notices.";
36
37 revision 2025-02-07 {
38   description
39     "Published as part of IEEE Std 802.1AXdz-202y.
40     The following reference statement identifies each referenced IEEE
41     Standard as updated by applicable amendments.";
42   reference
43     "IEEE Std 802.1AX-2020 Link Aggregation:
44     IEEE Std 802.1AXdz-202y.";
45 }
46
47 typedef key-group-ref {
48   type leafref {
49     path "/dotlax:linkagg/dotlax:key-group/dotlax:name";
50   }
51   description
52     "This type is used by aggregators and aggregation ports to
53     reference an entry in the key-group list.";
54 }
55
56 typedef agg-system-ref {
57   type leafref {
58     path "/dotlax:linkagg/dotlax:agg-system/dotlax:name";
59   }
60   description
61     "This type is used in the key-group list entries to
62     reference an entry in the agg-system list.";
63 }
64
65 augment "/if:interfaces/if:interface" {
66   when
67     "derived-from-or-self(if:type,'ianaif:ieee8023adLag') or "
68     + "derived-from-or-self(if:type,'ianaif:ieee8021axDrni')";
69   description
70     "Applies to interfaces representing a LAG or
71     (if supported) DRNI Intra-Relay Port.";
72 }
```

```
1  description
2    "Augment Interface with Aggregator parameters.";
3  container lag {
4    description
5      "Contains the configuration and status information that
6       allows an instance of an Aggregator to be managed.";
7    leaf key-group-name {
8      type key-group-ref;
9      mandatory true;
10     description
11       "Specifies the entry in the link-aggregation key-group
12        list to which this aggregator is assigned.";
13   }
14   leaf mac-address {
15     type ieee:mac-address;
16     config false;
17     description
18       "The MAC address assigned to the Aggregator.";
19     reference
20       "7.3.1.1.9 of IEEE Std 802.1AX";
21   }
22   container lacp {
23     config false;
24     description
25       "Contains aggregator LACP operational data.";
26     leaf actor-oper-key {
27       type uint16;
28       description
29         "The current operational value of the Key for the
30          Aggregator. The meaning of particular Key
31          values is of local significance.";
32       reference
33         "7.3.1.1.8 of IEEE Std 802.1AX";
34     }
35     leaf partner-system {
36       type ieee:mac-address;
37       description
38         "The value of the MAC address portion of the
39          System Identifier for the current
40          protocol partner of this Aggregator. A value
41          of zero indicates that there is no known partner.
42          If the aggregation is manually configured, this
43          System identifier value is assigned by the
44          local System.";
45       reference
46         "7.3.1.1.10 of IEEE Std 802.1AX";
47     }
48     leaf partner-system-priority {
49       type uint16;
50       description
51         "Indicates the priority value portion of the current
52          Partner's System Identifier. If the aggregation is
53          manually configured, this System Priority value is
54          assigned by the local System.";
55       reference
56         "7.3.1.1.11 of IEEE Std 802.1AX";
57     }
58     leaf partner-oper-key {
59       type uint16;
60       description
61         "The current operational value of the Key for the
62          Aggregators current protocol Partner. If the
63          aggregation is manually configured, this Key value
64          is assigned by the local System.";
65       reference
66         "7.3.1.1.12 of IEEE Std 802.1AX";
67     }
68     leaf aggregate-or-individual {
69       type boolean;
70       description
71         "Indicates whether the Aggregator represents an
72          Aggregate (TRUE) or an Individual link (FALSE).";
```

```

1      reference
2      "7.3.1.1.6 of IEEE Std 802.1AX";
3  }
4  }
5  container cscd {
6    if-feature "ax:cscd";
7    config false;
8    description
9      "Aggregator parameters obtained by the operation of LACP
10     supporting CSCD.";
11    leaf partner-port-algorithm {
12      type identityref {
13        base ax:distribution-algorithm;
14      }
15      description
16        "Operational value of the distribution algorithm in
17         use by the LACP Partner.";
18      reference
19        "7.3.1.1.40 of IEEE Std 802.1AX";
20    }
21    leaf partner-conv-service-digest {
22      type binary;
23      description
24        "The MD5 Digest of the admin-conv-service-map in use
25         by the LACP Partner.";
26      reference
27        "7.3.1.1.42, 6.6.3.1 of IEEE Std 802.1AX";
28    }
29    leaf partner-conv-link-digest {
30      type binary;
31      description
32        "The MD5 Digest of the admin-conv-link-map in use
33         by the LACP Partner.";
34      reference
35        "7.3.1.1.41, 6.6.3.1 of IEEE Std 802.1AX";
36    }
37    leaf discard-wrong-conversation {
38      type boolean;
39      description
40        "The operational value that determines whether an
41         Aggregator discards a frame that is collected
42         from an Aggregation Port that is different from the
43         Aggregation Port to which the Aggregator would
44         distribute a frame with the same Port Conversation
45         ID.";
46      reference
47        "7.3.1.1.37, 6.6 of IEEE Std 802.1AX";
48    }
49    leaf-list active-lag-links {
50      type uint16;
51      description
52        "A list, possibly empty, of the operational
53         link-number of each Aggregation Port active
54         (i.e. Collecting) on this Aggregator.";
55      reference
56        "7.3.1.1.43 of IEEE Std 802.1AX";
57    }
58  }
59 }
60 }
61
62 augment "/if:interfaces/if:interface/if:statistics" {
63   when '../dot1ax:lag' {
64     description
65       "Applies to aggregators.";
66   }
67   description
68     "Augment interface statistics with aggregator statistics.";
69   container lag-stats {
70     config false;
71     description
72       "Contains the set of stats associated with the

```

```
1     Aggregator.";
2     leaf unknown-protocol-frames {
3         type yang:counter64;
4         description
5             "A count of data frames discarded on reception by all
6             ports that are (or have been) members of the
7             aggregation, due to the detection of an unknown Slow
8             Protocols PDU (7.3.3.1.5)";
9         reference
10            "7.3.1.1.29 of IEEE Std 802.1AX";
11     }
12 }
13 }
14
15 augment "/if:interfaces/if:interface" {
16     description
17         "Augment interface model with Aggregation port
18         configuration nodes.";
19     container aggport {
20         presence
21             "When present, this interface supports Link Aggregation";
22         description
23             "Contains Aggregation Port configuration related nodes,
24             which provides the basic management controls necessary
25             to allow an instance of an Aggregation Port to be managed,
26             for the purposes of Link Aggregation.";
27         leaf key-group-name {
28             type key-group-ref;
29             mandatory true;
30             description
31                 "Specifies the entry in the link-aggregation key-group
32                 list to which this aggregation-port is assigned.";
33         }
34         leaf actor-port-number {
35             type uint16 {
36                 range "1..65535";
37             }
38             description
39                 "The port number assigned to the Aggregation Port.
40                 The port number is communicated in LACPDUs as the
41                 Actor Port.
42                 This leaf provides the ability to administratively
43                 override the initial value provided by the system.";
44             reference
45                 "7.3.2.1.14, 6.4.6 of IEEE Std 802.1AX";
46         }
47         leaf actor-port-priority {
48             type uint16;
49             default "0x8000";
50             description
51                 "The priority value assigned to this Aggregation Port.";
52             reference
53                 "7.3.2.1.15, 6.4.6 of IEEE Std 802.1AX";
54         }
55         leaf partner-admin-key {
56             type uint16 {
57                 range "1..65535";
58             }
59             description
60                 "The current administrative value of the Key for the
61                 protocol Partner. The assigned value is used, along
62                 with the value of port-partner-admin-system-priority,
63                 partner-admin-system, partner-admin-port, and
64                 partner-admin-port-priority, in order to achieve
65                 manually configured aggregation.
66                 The default value is the initial value of actor-port-
67                 number provided by the system.";
68             reference
69                 "7.3.2.1.10 of IEEE Std 802.1AX";
70         }
71         leaf partner-admin-port {
72             type uint16 {
```

```
1     range "1..65535";
2   }
3   description
4     "The current administrative value of the port number for
5     the protocol Partner. The assigned value is used, along
6     with the value of partner-admin-system-priority,
7     partner-admin-system, port-partner-admin-key, and
8     partner-admin-port-priority, in order to achieve
9     manually configured aggregation.
10    The default value is the initial value of actor-port-
11    number provided by the system.";
12   reference
13     "7.3.2.1.16 of IEEE Std 802.1AX";
14 }
15 leaf partner-admin-port-priority {
16   type uint16;
17   default "0";
18   description
19     "The current administrative value of the port priority
20     for the protocol Partner. The assigned value is used,
21     along with the value of partner-admin-system-priority,
22     partner-admin-system, partner-admin-key, and
23     partner-admin-port, in order to achieve manually
24     configured aggregation.";
25   reference
26     "7.3.2.1.18 of IEEE Std 802.1AX";
27 }
28 leaf partner-admin-state {
29   type ax:lacp-state {
30     bit lacp-activity {
31       position 1;
32       description
33         "When the LACP_Activity state is set to '0', the
34         transmission of LACPDUs is controlled by the
35         actor-admin-state.LACP_Activity.";
36     }
37     bit lacp-timeout {
38       position 2;
39       description
40         "When the LACP_Timeout is set to '0', LACPDUs
41         are transmitted at the Slow_Periodic_Time.";
42     }
43     bit aggregation {
44       position 3;
45       description
46         "When the Aggregation state is set to '0', this
47         port cannot be aggregated with any other port.";
48     }
49     bit synchronization {
50       position 4;
51       description
52         "When the Synchronization state is set to '0', this
53         port cannot become active.";
54     }
55   }
56   default "synchronization";
57   description
58     "Provides administrative control over the partner's
59     LACP_Activity, LACP_Timeout, Aggregation, and
60     Synchronization state when the partner's information is
61     unknown (i.e. no LACPDUs are received from the partner).";
62   reference
63     "7.3.2.1.22, 6.4.1, 6.4.2.3, 6.4.6 of IEEE Std 802.1AX";
64 }
65 leaf wtr-time {
66   type uint16 {
67     range "0..32767";
68   }
69   default "0";
70   description
71     "The wait-to-restore (WTR) period, in seconds, that
72     needs to elapse between an Aggregation Port on a LAG
```



```
1         coming up (Port_Operational becoming TRUE) and being
2         permitted to become active (transmitting and
3         receiving frames) on the LAG.";
4     reference
5         "7.3.2.1.30 of IEEE Std 802.1AX";
6 }
7 leaf wtr-revertive {
8     type boolean;
9     default "true";
10    description
11        "Controls revertive or non-revertive mode of operation.
12        When TRUE, the Aggregation Port can become active as
13        soon as the wait-to-restore timer expires regardless of
14        the state of other links in the LAG.
15        When FALSE, the Aggregation Port cannot become active
16        unless there are no other links that can become active
17        in the LAG. The default value is TRUE.";
18    reference
19        "7.3.2.1.31 of IEEE Std 802.1AX";
20 }
21 container lacp {
22     config false;
23     description
24         "Contains Aggregation port LACP operational related
25         nodes.";
26     leaf actor-lacp-version {
27         type uint8;
28         description
29             "The version number transmitted in LACPDUs on this
30             Aggregation Port";
31         reference
32             "7.3.2.1.33 of IEEE Std 802.1AX";
33     }
34     leaf actor-oper-key {
35         type uint16;
36         description
37             "The current operational value of the Key for the
38             Aggregation Port. The meaning of particular Key values
39             is of local significance.";
40         reference
41             "7.3.2.1.5 of IEEE Std 802.1AX";
42     }
43     leaf actor-oper-state {
44         type ax:lacp-state;
45         description
46             "The operational value of the Actor_State as
47             transmitted in LACPDUs.";
48         reference
49             "7.3.2.1.21, 6.4.1, 6.4.2.3, 6.4.6 of IEEE Std 802.1AX";
50     }
51     leaf partner-lacp-version {
52         type uint8;
53         description
54             "The version number in the LACPDU most recently
55             received on this Aggregation Port.";
56         reference
57             "7.3.2.1.34 of IEEE Std 802.1AX";
58     }
59     leaf partner-oper-system-priority {
60         type uint16;
61         description
62             "Indicates the operational value of priority associated
63             with the Partners System ID. The value of this
64             attribute can contain the manually configured value
65             carried in partner-admin-system-priority if there is
66             no protocol Partner.";
67         reference
68             "7.3.2.1.7 of IEEE Std 802.1AX";
69     }
70     leaf partner-oper-system {
71         type ieee:mac-address;
72         description
```

```
1      "Represents the MAC address portion of the Aggregation
2      Port's current protocol partner's System Identifier.
3      A value of zero indicates that there is no known
4      protocol Partner.
5      The value of this attribute can contain the manually
6      configured value carried in partner-admin-system if
7      there is no protocol Partner.";
8      reference
9      "7.3.2.1.9 of IEEE Std 802.1AX";
10     }
11     leaf partner-oper-key {
12         type uint16;
13         description
14         "The current operational value of the Key for the
15         protocol Partner. The value of this attribute can
16         contain the manually configured value carried in
17         partner-admin-key if there is no protocol Partner.";
18         reference
19         "7.3.2.1.11 of IEEE Std 802.1AX";
20     }
21     leaf partner-oper-port {
22         type uint16;
23         description
24         "The operational port number assigned by the
25         Aggregation Port's protocol Partner. The value of this
26         attribute can contain the administratively configured
27         value carried in partner-admin-port if there is no
28         protocol Partner.";
29         reference
30         "7.3.2.1.17 of IEEE Std 802.1AX";
31     }
32     leaf partner-oper-port-priority {
33         type uint16;
34         description
35         "The operational priority value assigned by the
36         Aggregation Port's protocol Partner. The value of this
37         attribute can contain the administratively configured
38         value carried in partner-admin-port-priority if there
39         is no protocol Partner.";
40         reference
41         "7.3.2.1.19 of IEEE Std 802.1AX";
42     }
43     leaf partner-oper-state {
44         type ax:lacp-state;
45         description
46         "The operational value of the partner's LACP state
47         derived from received LACPDUs or, when Defaulted is
48         true, from the partner-admin-state.";
49         reference
50         "7.3.2.1.23, 6.4.1, 6.4.2.3, 6.4.6 of IEEE Std 802.1AX";
51     }
52     leaf aggregate-or-individual {
53         type boolean;
54         description
55         "When true indicates the Aggregation Port can join a
56         LAG consisting of multiple Aggregation Ports.
57         When false, indicates that the Aggregation Port can
58         only operate as an Solitary link because the
59         Aggregation bit is false in either
60         actor-oper-port-state or partner-oper-port-state.";
61         reference
62         "7.3.2.1.24 of IEEE Std 802.1AX";
63     }
64     leaf wtr-waiting {
65         type boolean;
66         description
67         "Indicates the Aggregation Port is inhibited from
68         becoming active for an interval (determined by
69         wtr-time) after becoming operational or while
70         non-revertive operation is being enforced by the
71         Selection Logic.";
72         reference
```

```
1         "7.3.2.32 of IEEE Std 802.1AX";
2     }
3 }
4 container cscd {
5     if-feature "ax:cscd";
6     description
7         "Aggregation port parameters for support of CSCD.";
8     leaf admin-link-number {
9         type uint16;
10        description
11            "The Link_Number value for the Aggregation Port,
12             that is unique among all Aggregation Ports in the same
13             key group. This leaf provides the ability to
14             administratively override the initial value provided
15             by the system. A value of 0 is allowed, but
16             results in no frames being distributed to this
17             Aggregation Port.
18             The default value is the initial value of actor-port-
19             number provided by the system.";
20        reference
21            "7.3.2.1.27, 6.6.3.2 of IEEE Std 802.1AX";
22    }
23    leaf partner-link-number {
24        type uint16;
25        config false;
26        description
27            "The last received value of the Partner_Link_Number,
28             or zero if the Aggregation Port is using default
29             values for the Partner or the Partner LACP Version
30             is 1.";
31        reference
32            "7.3.2.1.29 of IEEE Std 802.1AX";
33    }
34    leaf link-number {
35        type uint16;
36        config false;
37        description
38            "The operational link number for this Aggregation Port.
39             The value is either the same as the admin-link-number,
40             or the corresponding value of the LACP partner.";
41        reference
42            "7.3.2.1.28 of IEEE Std 802.1AX";
43    }
44 }
45 }
46 }
47
48 augment "/if:interfaces/if:interface/if:statistics" {
49     when '../dot1ax:aggport' {
50         description
51             "Applies to aggregation ports.";
52     }
53     description
54         "Augment interface statistics with aggport statistics.";
55     container aggport-stats {
56         config false;
57         description
58             "Contains stats associated with the Aggregation Port.";
59         leaf lacp-pdu-rx {
60             type yang:counter64;
61             description
62                 "The number of valid LACPDUs received on this
63                 Aggregation Port.";
64             reference
65                 "7.3.3.1.2 of IEEE Std 802.1AX";
66         }
67         leaf marker-pdu-rx {
68             type yang:counter64;
69             description
70                 "The number of valid Marker PDUs received on this
71                 Aggregation Port.";
72             reference
```

```
1      "7.3.3.1.3 of IEEE Std 802.1AX";
2    }
3    leaf marker-response-pdu-rx {
4      type yang:counter64;
5      description
6        "The number of valid Marker Response PDUs received on
7        this Aggregation Port.";
8      reference
9        "7.3.3.1.4 of IEEE Std 802.1AX";
10   }
11   leaf unknown-rx {
12     type yang:counter64;
13     description
14       "The number of frames received that either:
15       a) Carry the Slow Protocols Ethernet Type value,
16       but contain an unknown PDU, or
17       b) Are addressed to the Slow Protocols group MAC
18       Address, but do not carry the Slow Protocols
19       Ethernet Type.";
20     reference
21       "7.3.3.1.5 of IEEE Std 802.1AX";
22   }
23   leaf illegal-rx {
24     type yang:counter64;
25     description
26       "The number of frames received that carry the Slow
27       Protocols Ethernet Type value, but contain a badly
28       formed PDU or an illegal value of Protocol Subtype.";
29     reference
30       "7.3.3.1.6 of IEEE Std 802.1AX";
31   }
32   leaf lacp-pdu-tx {
33     type yang:counter64;
34     description
35       "The number of LACPDUs transmitted on this
36       Aggregation Port.";
37     reference
38       "7.3.3.1.7 of IEEE Std 802.1AX";
39   }
40   leaf marker-pdu-tx {
41     type yang:counter64;
42     description
43       "The number of Marker PDUs transmitted on this
44       Aggregation Port.";
45     reference
46       "7.3.3.1.8 of IEEE Std 802.1AX";
47   }
48   leaf marker-response-pdu-tx {
49     type yang:counter64;
50     description
51       "The number of Marker Response PDUs transmitted on
52       this Aggregation Port.";
53     reference
54       "7.3.3.1.9 of IEEE Std 802.1AX";
55   }
56 }
57 }
58
59 container linkagg {
60   description
61     "Link Aaggregation System specific configuration nodes.";
62   list agg-system {
63     key "name";
64     description
65       "List of aggregation systems.";
66     leaf name {
67       type string;
68       description
69         "Name for the aggregation system.";
70     }
71     leaf actor-system {
72       type ieee:mac-address;
```

```

1      description
2          "The part of the System Identifier that is a globally
3          unique MAC address. This leaf provides the ability to
4          administratively override the initial value provided
5          by the system.";
6      reference
7          "7.3.1.1.4, 7.3.2.1.3 of IEEE Std 802.1AX";
8  }
9  leaf actor-system-priority {
10     type uint16;
11     default "0x8000";
12     description
13         "The part of the System Identifier that is the
14         priority of the system.";
15     reference
16         "7.3.1.1.5, 7.3.2.1.2 of IEEE Std 802.1AX";
17 }
18 }
19 list key-group {
20     key "name";
21     unique "actor-admin-key";
22     description
23         "List of key groups. A key group is the set of aggregators
24         and aggregation ports that share the same system priority,
25         system identifier, and aggregation key, and therefore can
26         potentially form a Link Aggregation Group. Each entry in
27         the key group list contains the parameters common to all
28         aggregation ports and/or aggregators in the key group.";
29     reference
30         "6.3.5, 6.4.12 of IEEE Std 802.1AX";
31     leaf name {
32         type string;
33         description
34             "Name for the key group.";
35     }
36     leaf actor-admin-key {
37         type uint16 {
38             range "1..65535";
39         }
40         mandatory true;
41         description
42             "The administrative value of the key used by the
43             Aggregators and Aggregation Ports in this key-group.";
44         reference
45             "7.3.1.1.7, 7.3.2.1.4 of IEEE Std 802.1AX";
46     }
47     leaf agg-system-name {
48         type agg-system-ref;
49         mandatory true;
50         description
51             "Specifies the aggregation system for this key
52             group.";
53     }
54     leaf actor-protocol-da {
55         type ieee:mac-address;
56         must ". = '01-80-c2-00-00-00' or . = '01-80-C2-00-00-00' or
57             . = '01-80-c2-00-00-02' or . = '01-80-C2-00-00-02' or
58             . = '01-80-c2-00-00-03' or . = '01-80-C2-00-00-03'" {
59             error-message "Invalid protocol address";
60         }
61         default "01-80-c2-00-00-02";
62         description
63             "A 6-octet read-write MAC Address value specifying the DA
64             to be used when sending Link Aggregation Control and
65             Marker PDUs. Valid addresses are the Nearest Customer
66             Bridge, Slow_Protocols_Multicast, and Nearest non-TPMR
67             Bridge group addresses. The default value
68             is the Slow_Protocols_Multicast address.";
69         reference
70             "7.3.2.2.1, 6.2.10.2 of IEEE Std 802.1AX";
71     }
72     leaf collector-max-delay {

```

```
1      type uint16;  
2      description  
3          "Specifies the maximum delay, in tens of microseconds,  
4          between receiving a frame from an Aggregator Port, and  
5          either delivering the frame to the Aggregator Client or  
6          discarding the frame. A value of zero means the delay  
7          is less than the minimum increment (< 10us).  
8          This leaf provides the ability to administratively  
9          override the initial value provided by the system.";  
10     reference  
11         "7.3.1.1.32, 6.2.3.1.1, B.3 of IEEE Std 802.1AX";  
12 }  
13 leaf actor-admin-state {  
14     type ax:lacp-state {  
15         bit lacp-activity {  
16             position 1;  
17             description  
18                 "Setting the LACP_Activity state to '0' means that the  
19                 transmission of LACPDUs is controlled by the value of  
20                 partner-oper-state.LACP_Activity.";  
21         }  
22         bit lacp-timeout {  
23             position 2;  
24             description  
25                 "Setting the LACP_Timeout to '0' means that actor uses  
26                 the Long_Timeout value, allowing the partner to transmit  
27                 LACPDUs at the Slow_Periodic_Time.";  
28         }  
29         bit aggregation {  
30             position 3;  
31             description  
32                 "Setting the Aggregation state to '0' prevents this  
33                 port from being aggregated with any other ports.";  
34         }  
35     }  
36     default "lacp-activity lacp-timeout aggregation";  
37     description  
38         "Provides administrative control over the values of the  
39         LACP_Activity, LACP_Timeout, and Aggregation state.";  
40     reference  
41         "7.3.2.1.20, 6.4.1, 6.4.2.3, 6.4.6 of IEEE Std 802.1AX";  
42 }  
43 leaf partner-admin-system {  
44     type ieee:mac-address;  
45     default "00-00-00-00-00-00";  
46     description  
47         "The administrative value of the MAC address portion of  
48         the Partner's System Identifier.  
49         The assigned value is used, along with the value of  
50         port-partner-admin-system, partner-admin-key,  
51         partner-admin-port, and partner-admin-port-priority,  
52         to achieve administratively configured Link  
53         Aggregation Groups with a partner that does not run  
54         LACP.";  
55     reference  
56         "7.3.2.1.8 of IEEE Std 802.1AX";  
57 }  
58 leaf partner-admin-system-priority {  
59     type uint16;  
60     default "0";  
61     description  
62         "The administrative value of priority portion of the  
63         the Partner's System Identifier. The assigned  
64         value is used, along with the value of  
65         port-partner-admin-system, partner-admin-key,  
66         partner-admin-port, and partner-admin-port-priority,  
67         to achieve administratively configured Link  
68         Aggregation Groups with a partner that does not run  
69         LACP.";  
70     reference  
71         "7.3.2.1.6 of IEEE Std 802.1AX";  
72 }
```

```
1 leaf port-algorithm {
2   type identityref {
3     base ax:distribution-algorithm;
4   }
5   default "ax:unspecified";
6   description
7     "Identifies the algorithm used by the Aggregator to
8      assign frames to a Port Conversation ID. Default is
9      the value for an unspecified distribution algorithm.
10     When the identity description specifies a 4 octet value,
11     this value will be used for cscd purposes, and included
12     in LACPDUs. Otherwise this leaf is used purely for
13     local selection of a distribution algorithm, and the
14     unspecified distribution algorithm is used for cscd
15     purposes.";
16   reference
17     "7.3.1.1.33, Table 8-1 of IEEE Std 802.1AX";
18 }
19 leaf-list lags {
20   type if:interface-ref;
21   config false;
22   description
23     "A list of the if:name of aggregators assigned to this
24     key group.";
25   reference
26     "linkagg:key-group";
27 }
28 leaf-list aggports {
29   type if:interface-ref;
30   config false;
31   description
32     "A list of the if:name of aggregation ports assigned to
33     this key group.";
34   reference
35     "linkagg:key-group";
36 }
37 container cscd {
38   if-feature "ax:cscd";
39   description
40     "Contains CSCD parameters that need to be consistent for
41     all aggregation ports and aggregators in the key group.";
42   container admin-conv-service-map {
43     if-feature "ax:sid-map";
44     description
45       "Data structure to map service identifiers to
46       conversation identifiers. Each entry consists of a
47       Conversation ID (CID) and a list of zero or more
48       Service Identifiers (SIDs) that map to it. An empty
49       list of SIDs means there are no SIDs that map to this
50       CID, and results in the same behavior as not having an
51       entry for this CID.";
45     reference
52       "7.3.1.1.36, 6.6.3.1 of IEEE Std 802.1AX";
53     uses ax:service-map;
54   }
55 }
56 leaf admin-conv-service-digest {
57   type binary;
58   config false;
59   description
60     "The MD5 Digest of the admin-conv-service-map. The
61     value is NULL (AAAAAAAAAAAAAAAAAAAAA== in base64) when
62     the distribution algorithm specified
63     specified by agg-port-algorithm does not use the
64     admin-conv-service-map.";
65   reference
66     "7.3.1.1.39, 6.6.3.1 of IEEE Std 802.1AX";
67 }
68 container admin-conv-link-map {
69   description
70     "Data structure to map a Conversation Identifier
71     (CID) to a Link Number. Each entry consists of a CID
72     and a list of link numbers that can potentially be
```

```

1         selected for that CID. An empty list of link-numbers
2         means that no links are selected for the CID.";
3     reference
4         "7.3.1.1.34, 6.6.3.1 of IEEE Std 802.1AX";
5     uses ax:link-map;
6 }
7 leaf admin-conv-link-digest {
8     type binary;
9     config false;
10    description
11        "The MD5 Digest of the admin-conv-link-map. The value
12        value is NULL (AAAAAAAAAAAAAAAAAAAAAA== in base64) when
13        the distribution algorithm specified
14        agg-port-algorithm does not use the
15        admin-conv-link-map.";
16    reference
17        "7.3.1.1.38, 6.6.3.1 of IEEE Std 802.1AX";
18 }
19 leaf admin-discard-wrong-conv {
20     if-feature "ax:dwc";
21     type enumeration {
22         enum force-true {
23             value 1;
24             description
25                 "Indicates that an Aggregator discards a
26                 frame that is collected from an Aggregation Port
27                 that is different from the Aggregation Port to
28                 which the Aggregator would distribute a frame
29                 with the same Port Conversation ID.";
30         }
31         enum force-false {
32             value 2;
33             description
34                 "Indicates that an Aggregator does not discard
35                 a frame that is collected from an Aggregation Port
36                 that is different from the Aggregation Port to
37                 which the Aggregator would distribute a frame with
38                 the same Port Conversation ID. This is the behavior
39                 of the Aggregator when DWC is not supported";
40         }
41         enum auto {
42             value 3;
43             description
44                 "Indicates that the Aggregator behaves as
45                 if the value was force-true only when the actor
46                 and partner agree on the algorithms (other than
47                 unspecified) and mapping tables used to map frames
48                 to Aggregation Ports, and behaves as if the value
49                 was force-false otherwise.";
50         }
51     }
52     default "force-false";
53     description
54         "Indicates whether an Aggregator discards a
55         frame that is collected from an Aggregation Port
56         that is different from the Aggregation Port to which
57         the Aggregator would distribute a frame with the
58         same Port Conversation ID.";
59     reference
60         "7.3.1.1.35, 6.6 of IEEE Std 802.1AX";
61 }
62 }
63 }
64 }
65 }
66
67

```

10.7.3 The *ieee802-dot1ax-drni* YANG module

```

69 module ieee802-dot1ax-drni {

```



```
1 yang-version 1.1;
2 namespace "urn:ieee:params:xml:ns:yang:ieee802-dotlax-drni";
3 prefix dotlax-drni;
4
5 import ieee802-dotlax-types {
6   prefix ax;
7 }
8 import ieee802-dotlax-linkagg {
9   prefix dotlax;
10 }
11 import ieee802-types {
12   prefix ieee;
13 }
14 import ietf-yang-types {
15   prefix yang;
16 }
17 import ietf-interfaces {
18   prefix if;
19 }
20
21 organization
22   "IEEE 802.1 Working Group";
23 contact
24   "WG-URL: http://www.ieee802.org/1/
25   WG-Email: stds-802-1-1@ieee.org
26
27   Contact: IEEE 802.1 Working Group Chair
28   Postal: C/O IEEE 802.1 Working Group
29           IEEE Standards Association
30           45 Hoes Lane
31           Piscataway, NJ 08854
32           USA
33   E-mail: stds-802-1-chairs@ieee.org";
34 description
35   "This YANG module describes the configuration model for a
36   Distributed Resilient Network Interface (DRNI) as specified
37   in 802.1AX.
38
39   Copyright (C) IEEE (2024).
40
41   This version of this YANG module is part of IEEE Std 802.1AX;
42   see the standard itself for full legal notices.";
43
44 revision 2025-02-07 {
45   description
46     "Published as part of IEEE Std 802.1AXdz-202y.
47     The following reference statement identifies each referenced IEEE
48     Standard as updated by applicable amendments.";
49   reference
50     "IEEE Std 802.1AX-2020 Link Aggregation:
51     IEEE Std 802.1AXdz-202y.";
52 }
53
54 typedef irp-state {
55   type bits {
56     bit reserved-1 {
57       position 1;
58       description
59         "Reserved for future use. It is set to 0 on
60         transmit and ignored on receipt.";
61     }
62     bit reserved-2 {
63       position 2;
64       description
65         "Reserved for future use. It is set to 0 on
66         transmit and ignored on receipt.";
67     }
68     bit short-timeout {
69       position 3;
70       description
71         "The Short_Timeout flag indicates the Timeout control value
72         in use by the DRCP Receive machine on this IRP. Short Timeout
```

```

1         is encoded as a 1; Long Timeout is encoded as a 0.";
2     }
3     bit synchronization {
4         position 4;
5         description
6             "When the Sync flag is TRUE (1), the DRCP Receive machine has
7             determined the Neighbor DRNI System has a compatible
8             configuration for forming a DRNI.";
9     }
10    bit irc-data {
11        position 5;
12        description
13            "When the IRC Data flag is TRUE (1), the transfer of Up
14            and Down frames is permitted on the IRC.";
15    }
16    bit drni {
17        position 6;
18        description
19            "The DRNI flag is TRUE (1) when this DRNI System is paired
20            with another DRNI System and FALSE (0) otherwise.";
21    }
22    bit defaulted {
23        position 7;
24        description
25            "When the Defaulted flag is TRUE (1), the DRCP Receive machine
26            is using default operational Neighbor information.
27            When FALSE (0), the operational Neighbor information
28            in use has been received in a DRCPDU.";
29    }
30    bit expired {
31        position 8;
32        description
33            "When the Expired flag is TRUE (1), the DRCP Receive machine
34            is in the EXPIRED state.";
35    }
36 }
37 description
38     "IRP state values as transmitted in DRCPDUs.
39     Administrative control over the values of Short_Timeout
40     and IRC_Data is allowed. The remaining bits are read-only.";
41 reference
42     "7.4.1.1.24, 9.6.2.3, Figure 9-13 of IEEE Std 802.1AX";
43 }
44
45 grouping drni-mask {
46     description
47         "Specifies the contents of a bit mask indexed by CID.";
48     leaf-list cid-str {
49         type string {
50             pattern '([0-9]{1}|[1-9]{1}[0-9]{1,2}|[1-3]{1}[0-9]{3})'
51             + '|40[0-8]{1}[0-9]{1}|409[0-5]{1})'
52             + '((-([0-9]{1}|[1-9]{1}[0-9]{1,2}|[1-3]{1}[0-9]{3})'
53             + '|40[0-8]{1}[0-9]{1}|409[0-5]{1})'
54             + '(%[0-9]{1,4})?)?)'
55             + '(',
56             + '([0-9]{1}|[1-9]{1}[0-9]{1,2}|[1-3]{1}[0-9]{3})'
57             + '|40[0-8]{1}[0-9]{1}|409[0-5]{1})'
58             + '((-([0-9]{1}|[1-9]{1}[0-9]{1,2}|[1-3]{1}[0-9]{3})'
59             + '|40[0-8]{1}[0-9]{1}|409[0-5]{1})'
60             + '(%[0-9]{1,4})?)?)'
61             + ')*';
62         }
63     default "0-4095";
64     description
65         "A comma separated list of single CID values (x),
66         ranges (x-y), or ranges with an increment (x-y%z).
67         For example, the string '0,10-13,100-110%4' sets the
68         bits in the mask corresponding to CIDs
69         0,10,11,12,13,100,104,108.
70         Some special cases are:
71         'x-y%z' where x+z>y is the same as 'x'.
72         'x-y%0' is the same as 'x'.
```

```
1      'x-y%1' is the same as 'x-y'.
2      'x-y%2' sets the bit for all even CIDs in the range
3      when x is even, and for all odd CIDs when x is odd.
4      Specified as a list of strings, so that long strings
5      can be broken into several shorter strings.";
6  }
7  leaf invert-mask {
8      type boolean;
9      default "false";
10     description
11         "When true the specified mask is inverted:
12         each zero replaced with a one,
13         and each one replaced with a zero.";
14 }
15 }
16
17 augment "/if:interfaces/if:interface/dot1ax:lag" {
18     description
19         "Augmentation parameters only for Aggregators with
20         DRNI enabled.";
21     container drni {
22         presence "When present, this Aggregator is enabled for DRNI";
23         description
24             "Aggregator parameters to support a Distributed
25             Resilient Network Interface";
26         leaf irp-name {
27             type if:interface-ref;
28             mandatory true;
29             description
30                 "Interface Name (if:name) of the Port supporting the
31                 Intra Relay Port (IRP) of this DRNI Gateway.";
32             reference
33                 "7.4.1.1.23 of IEEE Std 802.1AX";
34         }
35         leaf drni-aggregator-key {
36             type uint16 {
37                 range "1..65535";
38             }
39             description
40                 "The Aggregator Key value to be used by the Aggregator
41                 supporting this DRNI Gateway (and the Aggregation Ports
42                 assigned to this DRNI Gateway) when paired with a
43                 neighbor DRNI System via the IRC. The default value is
44                 the Aggregator's Actor_Admin_Aggregator_Key value.";
45             reference
46                 "7.4.1.1.15 of IEEE Std 802.1AX";
47         }
48         leaf drni-aggregator-system {
49             type ieee:mac-address;
50             default "00-00-00-00-00-00";
51             description
52                 "The Aggregator System value to be used by the
53                 Aggregator supporting this DRNI Gateway (and the
54                 Aggregation Ports assigned to this DRNI Gateway)
55                 when paired with a neighbor DRNI System via the
56                 Intra-Relay Connection (IRC).";
57             reference
58                 "7.4.1.1.13 of IEEE Std 802.1AX";
59         }
60         leaf drni-aggregator-system-priority {
61             type uint16;
62             default "0";
63             description
64                 "The Aggregator System Priority value to be used by the
65                 Aggregator supporting this DRNI Gateway (and the
66                 Aggregation Ports assigned to this DRNI Gateway) when
67                 paired with a neighbor DRNI System via the IRC.";
68             reference
69                 "7.4.1.1.14 of IEEE Std 802.1AX";
70         }
71         leaf drcp-protocol-da {
72             type ieee:mac-address;
```

```
1      must ". = '01-80-c2-00-00-00' or . = '01-80-C2-00-00-00' or
2          . = '01-80-c2-00-00-0e' or . = '01-80-C2-00-00-0E' or
3          . = '01-80-c2-00-00-03' or . = '01-80-C2-00-00-03'" {
4          error-message "Invalid protocol address";
5      }
6      default "01-80-c2-00-00-03";
7      description
8          "A 6-octet read-write MAC Address value specifying the
9          Destination Address for Distributed Relay Control PDUs
10         transmitted on the Intra-Relay Port. Valid addresses are
11         the Nearest Customer Bridge, Nearest Bridge, and
12         Nearest non-TPMR Bridge group addresses. The default
13         value is the Nearest Non-TPMR Bridge group
14         address.";
15      reference
16          "7.4.1.1.12, 9.6.1.1 of IEEE Std 802.1AX";
17  }
18  leaf home-admin-irp-state {
19      type irp-state {
20          bit short-timeout {
21              position 3;
22              description
23                  "The Short_Timeout flag indicates the Timeout control value
24                  in use by the DRCP Receive machine on this IRP. Short Timeout
25                  is encoded as a 1; Long Timeout is encoded as a 0.";
26          }
27          bit irc-data {
28              position 5;
29              description
30                  "When the IRC_Data flag is TRUE (1), the transfer of Up
31                  and Down frames is permitted on the IRC.";
32          }
33      }
34      default "short-timeout irc-data";
35      description
36          "Provides administrative control over the values of the
37          Short_Timeout and IRC-Data state.";
38      reference
39          "7.4.1.1.24, 9.6.2.3, Figure 9-13 of IEEE Std 802.1AX";
40  }
41  leaf home-oper-irp-state {
42      type irp-state;
43      config false;
44      description
45          "A string of 8 bits, corresponding to the current
46          operational value of IRP_State as transmitted in
47          DRCPDUs.";
48      reference
49          "7.4.1.1.25, 9.6.2.3, Figure 9-13 of IEEE Std 802.1AX";
50  }
51  leaf home-cscd-gateway-control {
52      type boolean;
53      default "true";
54      description
55          "When TRUE, allows the DRNI Gateway Port selection to
56          be based on the CSCD parameters that control the
57          Aggregator Port selection.";
58      reference
59          "7.4.1.1.16 of IEEE Std 802.1AX";
60  }
61  leaf home-dr-client-gateway-control {
62      type boolean;
63      default "true";
64      description
65          "When TRUE, allows the Distributed Relay Client to
66          determine whether to forward frames through the DRNI
67          Gateway Port.";
68      reference
69          "7.4.1.1.17 of IEEE Std 802.1AX";
70  }
71  leaf home-gateway-algorithm {
72      type identityref {
```

```

1      base ax:distribution-algorithm;
2    }
3    default "ax:unspecified";
4    description
5      "Identifies the algorithm used by the DRNI Gateway to
6      assign frames to a Gateway Conversation ID. 8.2 provides
7      the IEEE 802.1 OUI (00-80-C2) Gateway Algorithm
8      encodings. Default is the value for an unspecified
9      distribution algorithm.";
10   reference
11     "7.4.1.1.6 of IEEE Std 802.1AX";
12 }
13 container home-gateway-conv-service-map {
14   if-feature "ax:sid-map";
15   description
16     "Data structure to map service identifiers to
17     conversation identifiers. Each entry consists of a
18     Conversation ID (CID) and a list of zero or more Service
19     Identifiers (SIDs) that map to it. Frames with Service
20     IDs not contained in the map are not mapped to any
21     Gateway Conversation ID and are discarded.";
22   reference
23     "7.4.1.1.20, 6.6.3.1 of IEEE Std 802.1AX";
24   uses ax:service-map;
25 }
26 leaf home-gateway-conv-service-digest {
27   type binary;
28   config false;
29   description
30     "The MD5 Digest of the home-gateway-conv-service-map. The
31     value is NULL (AAAAAAAAAAAAAAAAAAAAAA== in base64) when
32     the distribution algorithm specified
33     by agg-port-algorithm does not use the
34     home-gateway-conv-service-map.";
35   reference
36     "7.4.1.1.21 of IEEE Std 802.1AX";
37 }
38 container gateway-enable-mask {
39   description
40     "A vector of Boolean values, with one value for each
41     possible Gateway Conversation ID. A 1 indicates that
42     frames associated with that Gateway Conversation ID
43     are allowed to pass through this Gateway Port, and a
44     0 indicates that such frames are not allowed to pass.
45     Default value is all bits set to 1.";
46   reference
47     "7.4.1.1.18, 9.5.3.5, 9.6.5 of IEEE Std 802.1AX";
48   uses drni-mask;
49 }
50 container gateway-preference-mask {
51   description
52     "A vector of Boolean values, with one value for each
53     possible Gateway Conversation ID. A 1 indicates that
54     this Gateway Port is the preferred Gateway when both
55     DRNI Gateways have the Gateway Conversation ID enabled
56     in the gateway-available-mask, and a 0 indicates that
57     it is not preferred.
58     Default value is all bits set to 1.";
59   reference
60     "7.4.1.1.19, 9.5.3.5, 9.6.5 of IEEE Std 802.1AX";
61   uses drni-mask;
62 }
63 container gateway-available-mask {
64   config false;
65   description
66     "A vector of Boolean values, with one value for each
67     possible Gateway Conversation ID. A 1 indicates that
68     this Gateway Port is eligible to be selected to pass
69     that Gateway Conversation ID, and a 0 indicates that
70     it is not eligible.";
71   reference
72     "7.4.1.1.22, 9.5.3.5, 9.6.5 of IEEE Std 802.1AX";

```

```

1      uses drni-mask;
2  }
3  container neighbor {
4      config false;
5      description
6          "Operational values for the DRNI neighbor obtained
7           from DRCPDUs.";
8      leaf oper-irp-state {
9          type irp-state;
10         description
11             "A string of 8 bits, corresponding to the current
12             operational value of IRP_State as transmitted in
13             DRCPDUs.";
14         reference
15             "7.4.1.1.25, 9.6.2.3, Figure 9-13 of IEEE Std 802.1AX";
16     }
17     leaf system {
18         type ieee:mac-address;
19         description
20             "The MAC Address portion of the System Identifier of
21             the Neighbor DRNI System (connected via the
22             Intra-Relay Port). ";
23         reference
24             "7.4.1.1.29 of IEEE Std 802.1AX";
25     }
26     leaf system-priority {
27         type uint16;
28         description
29             "The priority portion of the System Identifier of the
30             Neighbor DRNI System (connected via the Intra-Relay
31             Port).";
32         reference
33             "7.4.1.1.30 of IEEE Std 802.1AX";
34     }
35     leaf drni-key {
36         type uint16;
37         description
38             "The DRNI key value received from the Neighbor DRNI
39             System (connected via the IntraRelay Port).";
40         reference
41             "7.4.1.1.31 of IEEE Std 802.1AX";
42     }
43     leaf aggregator-algorithm {
44         type identityref {
45             base ax:distribution-algorithm;
46         }
47         description
48             "The Port algorithm used by the Neighbor Aggregator to
49             assign frames to Port Conversation IDs.";
50         reference
51             "7.4.1.1.33 of IEEE Std 802.1AX";
52     }
53     leaf aggregator-conv-service-digest {
54         type binary;
55         description
56             "The MD5 Digest of the Neighbor Aggregator's
57             Admin_Conv_Service_Map. Obtained from the Home
58             Aggregator State TLV last received from the Neighbor
59             DRNI System.";
60         reference
61             "7.4.1.1.34 of IEEE Std 802.1AX";
62     }
63     leaf aggregator-conv-link-digest {
64         type binary;
65         description
66             "The MD5 Digest of the Neighbor Aggregator's
67             Admin_Conv_Link_Map. Obtained from the Home Aggregator
68             State TLV (9.6.2.4) last received from the Neighbor
69             DRNI System.";
70         reference
71             "7.4.1.1.35 of IEEE Std 802.1AX";
72     }

```

```
1 leaf partner-system-priority {
2   type uint16;
3   description
4     "The priority portion of the System Identifier of the
5     Neighbor Aggregator's Partner.";
6   reference
7     "7.4.1.1.36 of IEEE Std 802.1AX";
8 }
9 leaf partner-system {
10  type ieee:mac-address;
11  description
12    "The MAC Address portion of the System Identifier of
13    the Neighbor Aggregator's Partner.";
14  reference
15    "7.4.1.1.37 of IEEE Std 802.1AX";
16 }
17 leaf partner-aggregator-key {
18  type uint16;
19  description
20    "The operational key value of the Neighbor
21    Aggregator's Partner.";
22  reference
23    "7.4.1.1.38 of IEEE Std 802.1AX";
24 }
25 leaf cscd-state {
26  type bits {
27    bit reserved-1 {
28      position 0;
29      description
30        "Bit 1 is reserved for future use. It is set to 0
31        and ignored on receipt.";
32    }
33    bit reserved-2 {
34      position 1;
35      description
36        "Bit 2 is reserved for future use. It is set to 0
37        and ignored on receipt.";
38    }
39    bit reserved-3 {
40      position 2;
41      description
42        "Bit 3 is reserved for future use. It is set to 0
43        and ignored on receipt.";
44    }
45    bit cscd_gateway_control {
46      position 3;
47      description
48        "CSCD_Gateway_Control is encoded in bit 4. When
49        this flag is TRUE, the DRNI Gateway is configured
50        to minimize forwarding data frames on the IRC by
51        selecting the DRNI Gateway and Aggregator Ports
52        for forwarding any given Conversation ID to be in
53        the same DRNI System.";
54    }
55    bit discard_wrong_conversation {
56      position 4;
57      description
58        "Discard_Wrong_Conversation is encoded in bit 5.
59        The Aggregator's Discard_Wrong_Conversation
60        value.";
61    }
62    bit differ_conv_link_digests {
63      position 5;
64      description
65        "Differ_Conv_Link_Digests is encoded in bit 6.
66        This flag is TRUE when the Aggregator's
67        Actor_Conv_Link_Digest matches the Aggregator's
68        Partner_Conv_Link_Digest.";
69    }
70    bit differ_conv_service_digests {
71      position 6;
72      description
```

```
1         "Differ_Conv_Service_Digests is encoded in bit 7.  
2         This flag is TRUE when the Aggregator's  
3         Actor_Conv_Service_Digest matches the Aggregator's  
4         Partner_Conv_Service_Digest.";  
5     }  
6     bit differ_port_algorithms {  
7         position 7;  
8         description  
9             "Differ_Port_Algorithms is encoded in bit 8. The  
10            Aggregator's differPortAlgorithms flag is TRUE  
11            when the Aggregator's Actor_Port_Algorithm matches  
12            the Aggregator's Partner_Port_Algorithm.";  
13     }  
14 }  
15 description  
16     "8 bits, corresponding to the Aggregator_CSCD_State  
17     in the Neighbor_Aggregator_State variable. The first  
18     three bits (the least significant bits of CSCD_State)  
19     are reserved; the fourth bit corresponds to the  
20     Neighbor's value for Home_Admin_CSCD_Gateway_Control;  
21     the fifth bit corresponds to the Neighbor Aggregator's  
22     operational value for Discard_Wrong_Conversation; and  
23     the sixth, seventh, and eighth bits correspond to the  
24     Neighbor Aggregator's operational value for  
25     differConvLinkDigests, differConvServiceDigests, and  
26     differPortAlgorithms, respectively, (the most  
27     significant bits of CSCD_State).";  
28 reference  
29     "7.4.1.1.39 of IEEE Std 802.1AX";  
30 }  
31 leaf-list active-links {  
32     type uint16;  
33     description  
34         "A list of the operational Link_Numbers of Aggregation  
35         Ports that are currently active (i.e., collecting) on  
36         the Neighbor's Aggregator. An empty list indicates that  
37         there are no Aggregation Ports active. Each integer  
38         value in the list carries an aAggPortOperLinkNumber  
39         attribute value.";  
40     reference  
41         "7.4.1.1.40 of IEEE Std 802.1AX";  
42 }  
43 leaf gateway-algorithm {  
44     type identityref {  
45         base ax:distribution-algorithm;  
46     }  
47     description  
48         "The gateway algorithm used by the Neighbor DRNI  
49         Gateway to assign frames to Gateway Conversation IDs.";  
50     reference  
51         "7.4.1.1.41 of IEEE Std 802.1AX";  
52 }  
53 leaf gateway-conv-service-digest {  
54     type binary;  
55     description  
56         "The MD5 Digest of the Neighbor DRNI Gateway's  
57         the Home_Admin_Gateway_Conv_Service_Map. Obtained  
58         from Gateway_Conv_Service_Digest in the  
59         Neighbor_Gateway_State TLV last received from the  
60         Neighbor DRNI System.";  
61     reference  
62         "7.4.1.1.42 of IEEE Std 802.1AX";  
63 }  
64 container gateway-available-mask {  
65     description  
66         "A vector of Boolean values, with one value for each  
67         possible Gateway Conversation ID. A 1 indicates that  
68         the Neighbor DRNI Gateway Port is eligible to be  
69         selected to pass that Gateway Conversation ID, and  
70         a 0 indicates that it is not eligible.";  
71     reference  
72         "7.4.1.1.43 of IEEE Std 802.1AX";
```



```
1      uses drni-mask;
2    }
3    container gateway-preference-mask {
4      description
5        "A vector of Boolean values, with one value for each
6        possible Gateway Conversation ID. A 1 indicates that
7        the Neighbor DRNI Gateway Port is the preferred
8        Gateway when both DRNI Gateways have the Gateway
9        Conversation ID enabled in the gateway-available-mask,
10       and a 0 indicates that it is not preferred.";
11      reference
12        "7.4.1.1.44 of IEEE Std 802.1AX";
13      uses drni-mask;
14    }
15  }
16 }
17 }
18
19 augment "/if:interfaces/if:interface/if:statistics" {
20   when '../dotlax:lag/dotlax-drni:drni' {
21     description
22       "Applies to aggregators with DRNI present.";
23   }
24   description
25     "Augment interface statistics with DRNI statistics.";
26   container drni-stats {
27     description
28       "Contains DRNI specific statistics.";
29     leaf drcpdus-rx {
30       type yang:counter64;
31       description
32         "The number of valid DRCPDUs received on this
33         Intra-Relay Port.";
34       reference
35         "7.4.1.1.45 of IEEE Std 802.1AX";
36     }
37     leaf illegal-rx {
38       type yang:counter64;
39       description
40         "The number of frames received on this Intra-Relay
41         Port that carry the DRCP EtherType value,
42         but contain a badly formed PDU.";
43       reference
44         "7.4.1.1.46, 9.6.1.4 of IEEE Std 802.1AX";
45     }
46     leaf drcpdus-tx {
47       type yang:counter64;
48       description
49         "The number of valid DRCPDUs transmitted on this
50         Intra-Relay Port.";
51       reference
52         "7.4.1.1.47 of IEEE Std 802.1AX";
53     }
54   }
55 }
56 }
57
58
59
```

Annex A

(normative)

Protocol implementation conformance statement (PICS) proforma⁶

Insert the following rows at the end of the table in A.2.1

A.2.1 Major capabilities/options

Item	Feature	Status	References	Support
YANG	Does the implementation support management operations using YANG modules?	O	10.6	Yes [] No []
YANG modules	Is the ieee802-dot1ax-types module supported?	YANG:M	10.7.1	N/A [] Yes []
	Is the ieee802-dot1ax-linkagg module supported?	YANG:M	10.7.2	N/A [] Yes []
	Is the ieee802-dot1ax-drni module supported?	DRNI AND YANG:M	10.7.3	N/A [] Yes []

6

⁶ Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

¹ **Annex G**

² (informative)

³ **Bibliography**

⁴ *Insert the following two bibliography entries after [B8] in Annex G:*

⁵

⁶ [B9] IETF RFC 7803, Changing the Registration Policy for the NETCONF Capability URNs Registry,
⁷ February 2016.

⁸ [B10] IETF RFC 8040, RESTCONF Protocol, January 2017.

⁹ [B11] OMG Unified Modeling Language (OMG UML), Version 2.5, March 2015.⁷

⁷OMG documents are available from the Object Management Group (<https://www.omg.org/>).